Personal Music Architect

Purpose of the project

To build a personal AI-powered music agent capable of interacting with a Spotify library using natural language.

- Intelligent querying of music libraries
- Dynamic creation and modification of playlists.
- Advanced analysis of listening behavior
- Semantic retrieval through vector search

Architecture

Intelligent Orchestration

The system uses a layered architecture that separates conversation management, graph workflow orchestration, and access to external data sources.

Services integration

It combines powerful APIs such as the Spotify Web API, OpenAI Embeddings, and the Pinecone vector database for semantic reasoning.

Core system compounds

Session Manager

Orchestrates the main entry point, detects intent, and maintains conversation state and context.

Graph Engine

Graph-based orchestration to determine precise actions based on the current musical state.

Repository Layer

It abstracts access to SQLite, acting as the source of truth for all musical knowledge.

Spotify Integration

Deep integration through the Spotipy client and OAuth authentication, enabling full control of the musical experience.

- Incremental synchronization
- Playlist management (CRUD)
- Access to recently played tracks

Spotify
AD Studio

Semantic Layer and Anchors

Emotional Anchors (RAG)

The system uses special playlists with the ANCHOR prefix to create semantic clusters.

This enables searches based on vibes or concepts, such as:
"Play music with the same feeling as my relax playlist."

Implemented using OpenAI Embeddings and vector search in Pinecone.

System Execution Flow

User Request

Intent detection

Graph Execution

Spotify API

Update DB/Vector

Agent Response

Implemented Capabilities

Dynamic Playlists

Generation and modification of music collections using natural language.

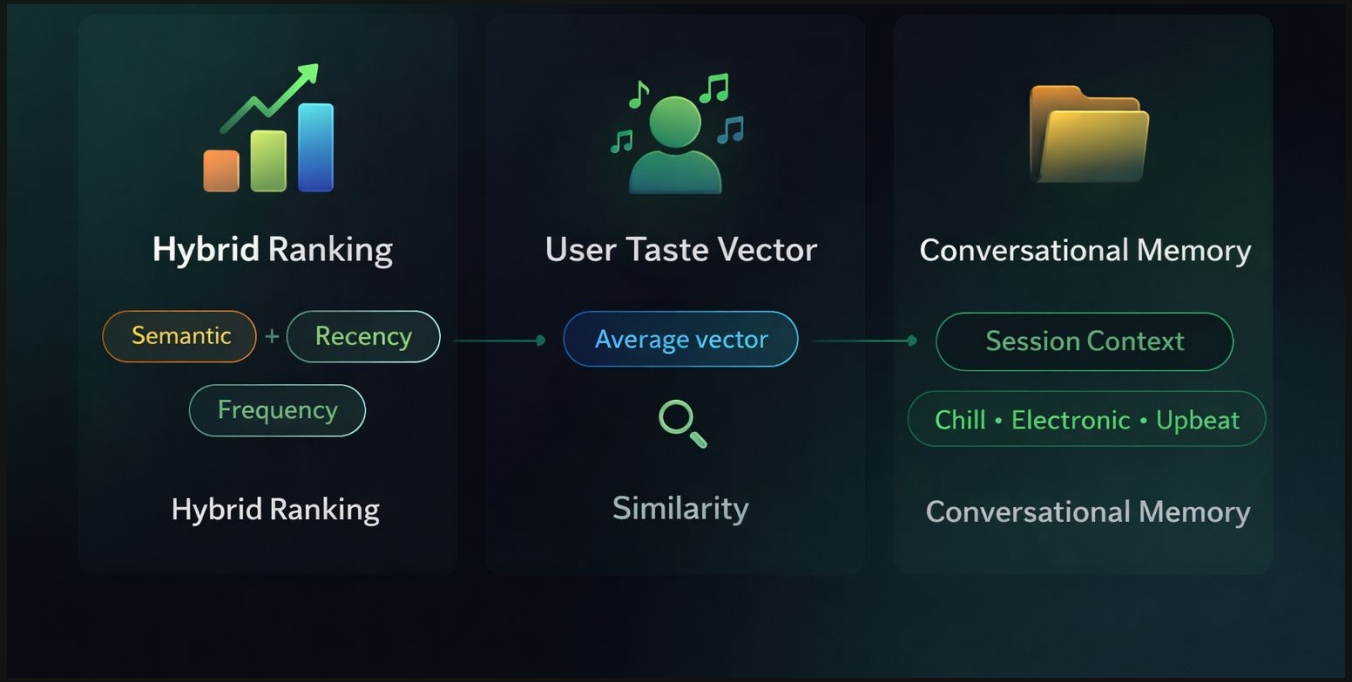
Listening Analysis

Discovery based on conceptual similarity and vector search (RAG).

Technical Challenges Overcome

- Data Consistency: Efficient local synchronization with the Spotify API.
- Multi-Layer Integration: Alignment of relational models with vector indexes.
- Conversation Context: Management of intent transitions and session memory.
- Ingestion Efficiency: Change detection without reprocessing the entire dataset.

Future Improvements



Hybrid-ranking (Semantic + recency + frequency)

User Taste Vector : intelligent musical profile (“recommend something I might like”)

Deep conversational music memory.

Questions?

Personal Music Architect Project

github.com/personal-music-architect

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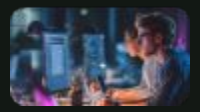
Image Sources



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Source: uxgirl.com

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Source: artists.sonomusic.co

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